

Run report

2026-09-11T160419Z

Configuration

Parameter	Value	Parameter	Value
batch_size	1,024	n_samples	1,600,000
dim	12	dataset	jets
seed	1,851,207,501	data_seed	42
hidden_units	128	n_layers	3
n_epochs	100	n_disc_steps	5
lr_g	3.000×10^{-5}	lr_d	1.000×10^{-4}
lambda_dispersion	0.01500	mmd_subsample	16,384
mmd_test	2.573×10^{-4}	best_epoch	91
compile_cache_warm	True		
variables	m [GeV], $\ln \rho$, $\lambda_{0.5}^1$, w , λ_2^1 , z_g , $\tau_{21}^{(\beta=1)}$, M , n_{ch} , f_{ch} , p_T^D , q		
mmd_sigmas_detector	$3.107 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		
mmd_sigmas_particle	$3.027 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		

Timing

Phase	Seconds	Share (%)	Detail
train	40.643	75.9	train
transfer	5.714		train
compile	3.910		train
epochs	29.729		train
select	1.080		train
particle_mmd	0.403	0.8	train
save	0.352	0.7	train
data	1.486	2.8	cache hit – load
load	1.404	2.6	load
plots	9.273	17.3	load
evaluate	0.004	0.0	load
total	53.565	100.0	

Detector-level metrics

Distance to Data. Improvements are relative to Simulation.

Wasserstein [$\times 10^3$]

Observable	Sim.	IBU	IBU impr. (%)	RAN	RAN impr. (%)
Mass and hard scale (IRC-safe kinematics)					
m [GeV]	208.9	138.8	33.6	9.340	95.5
$\ln \rho$	227.9	20.71	90.9	11.60	94.9
Continuous angularities (IRC-safe jet shapes)					
$\lambda_{0.5}^1$	262.6	12.84	95.1	7.440	97.2
w	265.6	9.978	96.2	6.082	97.7
λ_2^1	255.4	9.215	96.4	6.440	97.5
Splitting and 2-prong substructure					
$z_g^\dagger$	60.45	60.45	0.0	17.11	71.7
$\tau_{21}^{(\beta=1)}$	121.8	70.10	42.4	18.78	84.6
Hadronization, multiplicity and fragmentation (IRC-unsafe)					
M	224.2	177.3	20.9	20.78	90.7
n_{ch}	220.8	172.3	22.0	22.69	89.7
f_{ch}	52.86	40.94	22.6	7.496	85.8
p_T^D	256.8	30.80	88.0	16.82	93.4
q	79.67	17.51	78.0	5.595	93.0

[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

JS divergence [$\times 10^3$]

Observable	Sim.	IBU	IBU impr. (%)	RAN	RAN impr. (%)
Mass and hard scale (IRC-safe kinematics)					
m [GeV]	6.762	3.666	45.8	0.06873	99.0
$\ln \rho$	8.383	0.4017	95.2	0.1500	98.2
Continuous angularities (IRC-safe jet shapes)					
$\lambda_{0.5}^1$	9.359	0.4714	95.0	0.1141	98.8
w	9.139	0.4527	95.0	0.1155	98.7
λ_2^1	7.776	0.1728	97.8	0.1204	98.5
Splitting and 2-prong substructure					
$z_g^\dagger$	0.6723	0.6723	0.0	0.1136	83.1
$\tau_{21}^{(\beta=1)}$	3.372	1.934	42.7	0.1702	95.0
Hadronization, multiplicity and fragmentation (IRC-unsafe)					
M	9.640	7.633	20.8	0.1704	98.2
n_{ch}	7.512	6.314	16.0	0.1431	98.1
f_{ch}	1.033	0.7960	22.9	0.09115	91.2
p_T^D	8.223	1.311	84.1	0.1381	98.3
q	1.576	0.2512	84.1	0.08142	94.8

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VLC divergence [$\times 10^3$]

Observable	Sim.	IBU	IBU impr. (%)	RAN	RAN impr. (%)
Mass and hard scale (IRC-safe kinematics)					
m [GeV]	26.93	14.62	45.7	0.2578	99.0
$\ln \rho$	33.31	1.576	95.3	0.5797	98.3
Continuous angularities (IRC-safe jet shapes)					
$\lambda_{0.5}^1$	37.19	1.877	95.0	0.4506	98.8
w	36.32	1.800	95.0	0.4541	98.7
λ_2^1	30.90	0.6583	97.9	0.4426	98.6
Splitting and 2-prong substructure					
$z_g^\dagger$	2.683	2.683	0.0	0.4545	83.1
$\tau_{21}^{(\beta=1)}$	13.44	7.712	42.6	0.6761	95.0
Hadronization, multiplicity and fragmentation (IRC-unsafe)					
M	38.25	30.31	20.8	0.6497	98.3
n_{ch}	29.81	25.07	15.9	0.5460	98.2
f_{ch}	4.125	3.177	23.0	0.3645	91.2
p_T^D	32.65	5.211	84.0	0.5479	98.3
q	6.261	0.9940	84.1	0.3147	95.0
[†] IBU's purity binning produced a single bin, so IBU failed to unfold.					

Particle-level metrics

Distance to Truth. Improvements are relative to Generation.

Wasserstein [$\times 10^3$]

Observable	Gen	IBU	IBU impr. (%)	RAN	RAN impr. (%)
Mass and hard scale (IRC-safe kinematics)					
m [GeV]	111.7	94.07	15.8	86.32	22.8
$\ln \rho$	253.1	32.57	87.1	17.29	93.2
Continuous angularities (IRC-safe jet shapes)					
$\lambda_{0.5}^1$	303.0	18.71	93.8	6.176	98.0
w	284.1	12.29	95.7	3.450	98.8
λ_2^1	253.8	11.13	95.6	3.172	98.8
Splitting and 2-prong substructure					
$z_g^\dagger$	113.4	113.4	0.0	24.69	78.2
$\tau_{21}^{(\beta=1)}$	259.2	150.5	41.9	11.62	95.5
Hadronization, multiplicity and fragmentation (IRC-unsafe)					
M	256.8	204.8	20.2	24.19	90.6
n_{ch}	234.6	188.6	19.6	24.70	89.5
f_{ch}	45.78	45.33	1.0	5.079	88.9
p_T^D	236.3	55.36	76.6	17.61	92.5
q	79.37	18.15	77.1	5.307	93.3

[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

JS divergence [$\times 10^3$]

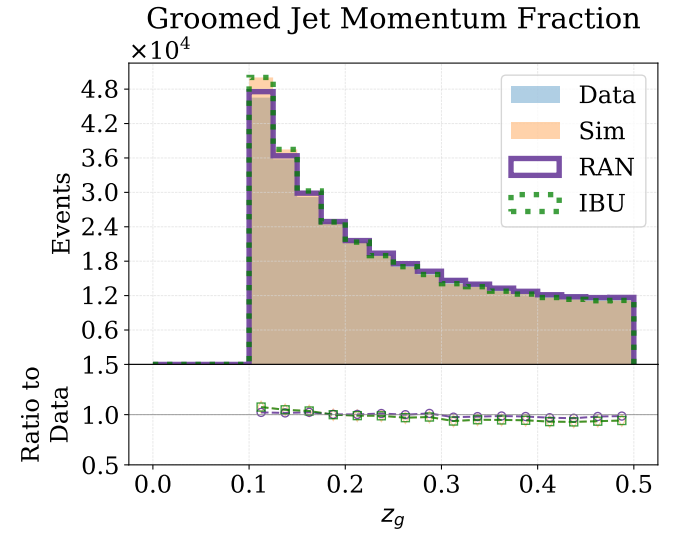
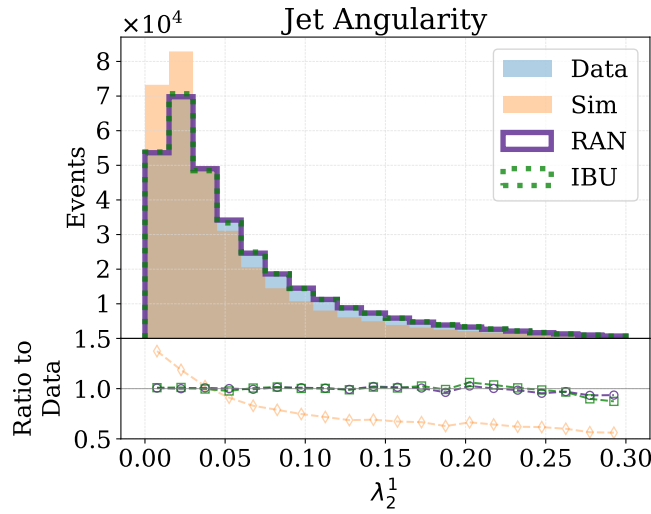
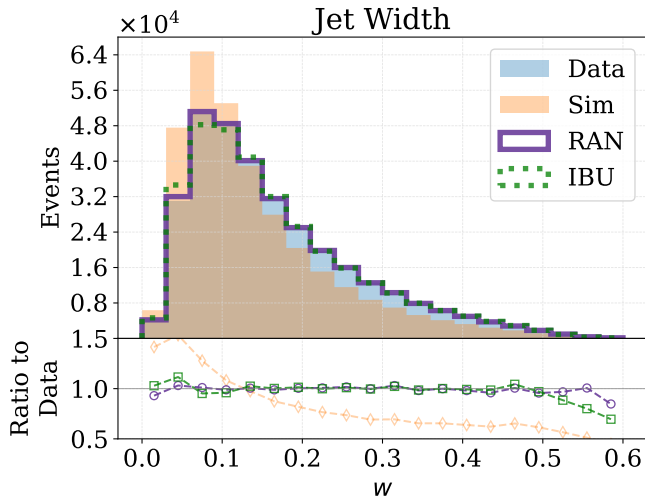
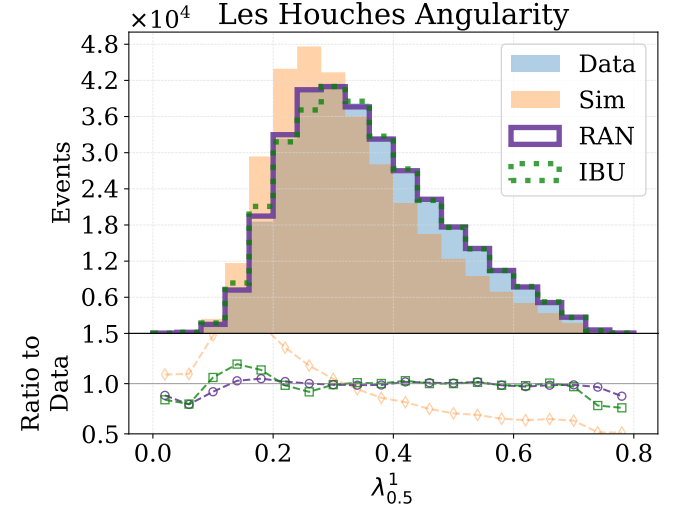
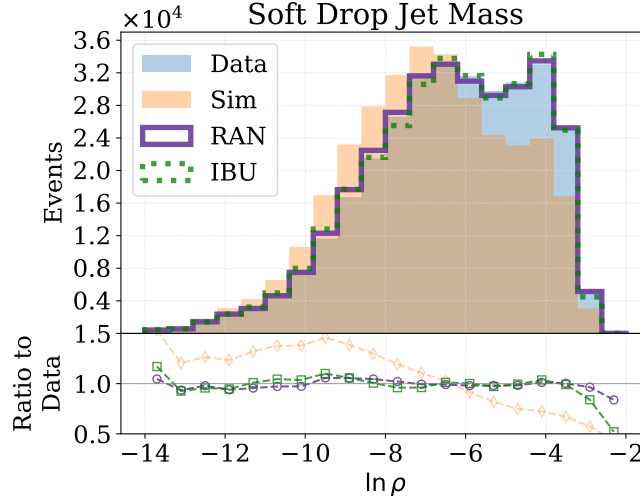
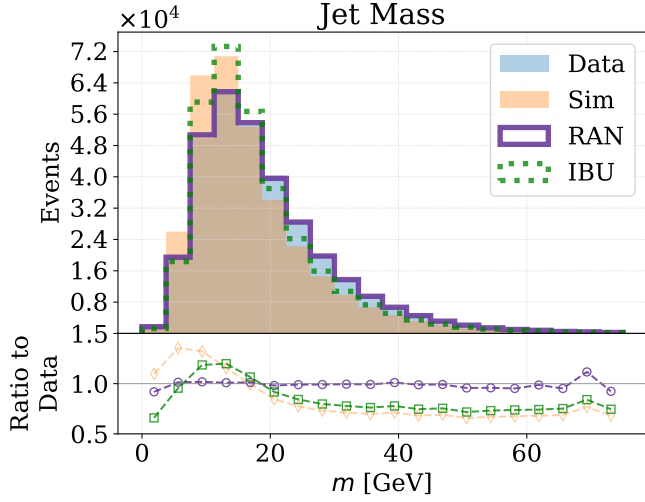
Observable	Gen	IBU	IBU impr. (%)	RAN	RAN impr. (%)
Mass and hard scale (IRC-safe kinematics)					
m [GeV]	1.963	11.86	−504.3	1.596	18.7
$\ln \rho$	9.878	1.731	82.5	0.1995	98.0
Continuous angularities (IRC-safe jet shapes)					
$\lambda_{0.5}^1$	12.34	2.009	83.7	0.1552	98.7
w	11.39	1.465	87.1	0.1688	98.5
λ_2^1	8.824	0.8335	90.6	0.1161	98.7
Splitting and 2-prong substructure					
$z_g^\dagger$	2.041	2.041	0.0	0.1972	90.3
$\tau_{21}^{(\beta=1)}$	10.50	6.577	37.4	0.1412	98.7
Hadronization, multiplicity and fragmentation (IRC-unsafe)					
M	7.842	12.83	−63.7	0.2621	96.7
n_{ch}	6.447	9.773	−51.6	0.1872	97.1
f_{ch}	1.025	0.9435	7.9	0.1121	89.1
p_T^D	8.342	4.029	51.7	0.2841	96.6
q	2.027	0.9958	50.9	0.1102	94.6

[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

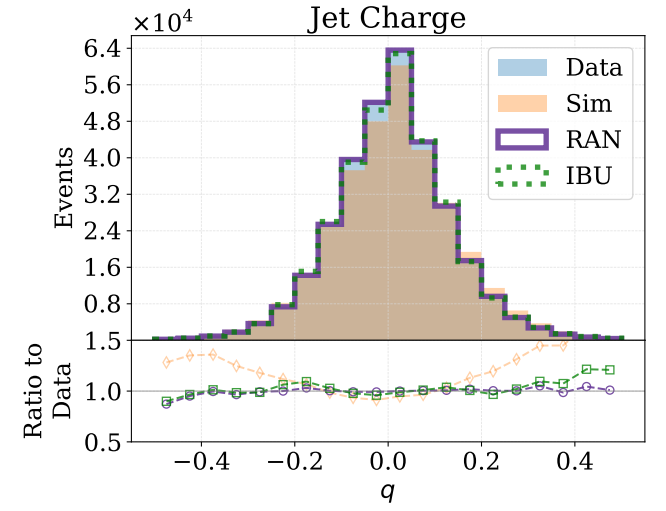
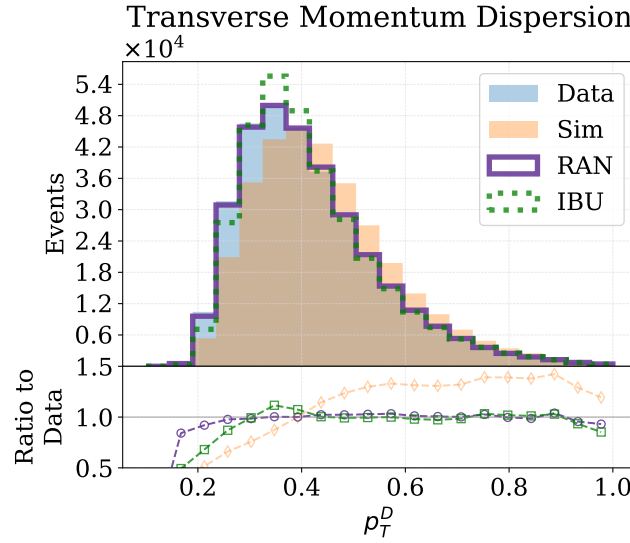
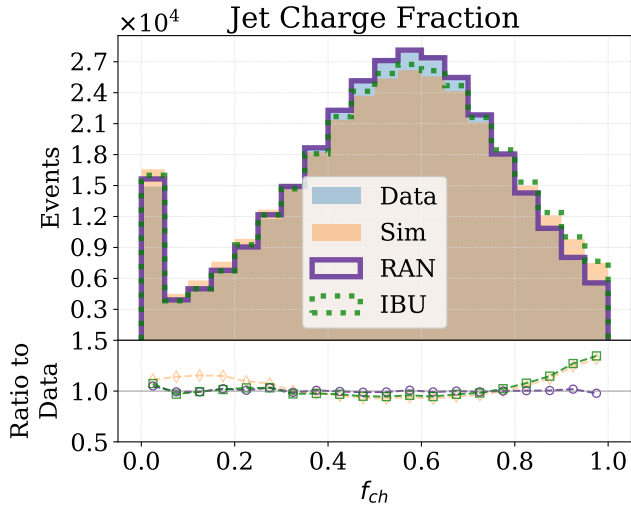
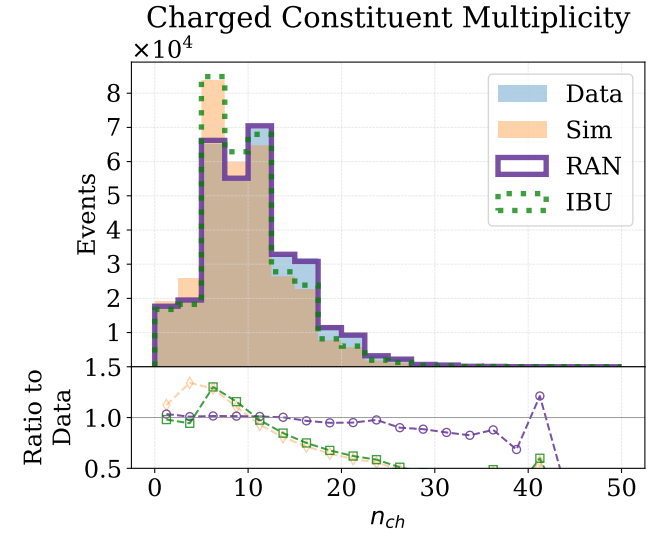
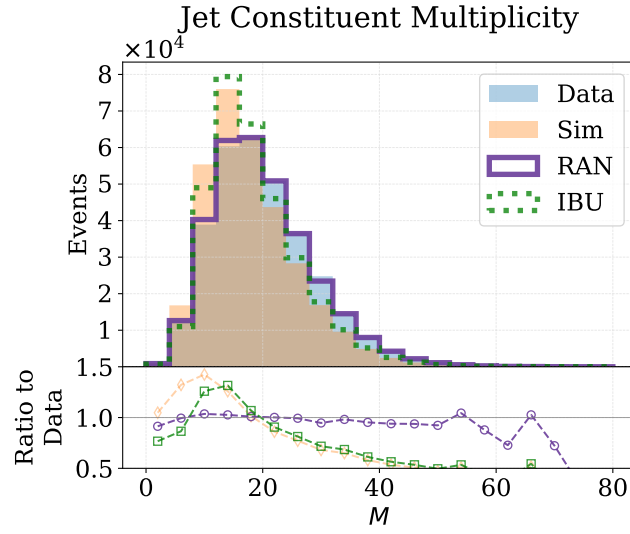
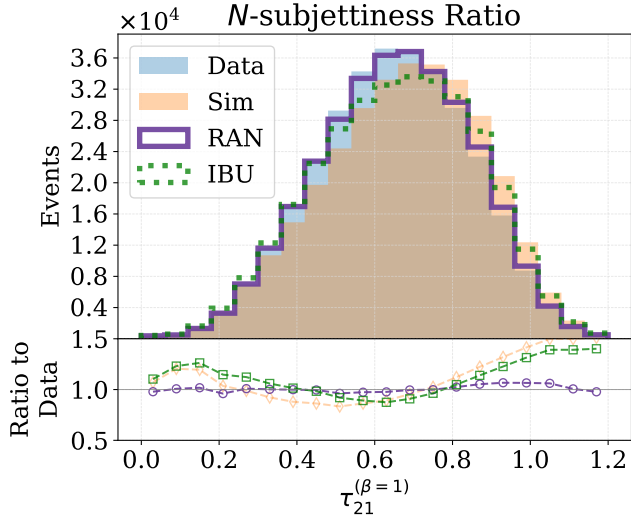
VLC divergence [$\times 10^3$]

Observable	Gen	IBU	IBU impr. (%)	RAN	RAN impr. (%)
Mass and hard scale (IRC-safe kinematics)					
m [GeV]	7.809	45.93	−488.2	6.347	18.7
$\ln \rho$	39.27	6.889	82.5	0.7830	98.0
Continuous angularities (IRC-safe jet shapes)					
$\lambda_{0.5}^1$	48.98	8.014	83.6	0.6153	98.7
w	45.23	5.837	87.1	0.6574	98.5
λ_2^1	35.08	3.315	90.5	0.4475	98.7
Splitting and 2-prong substructure					
$z_g^\dagger$	8.086	8.086	0.0	0.7875	90.3
$\tau_{21}^{(\beta=1)}$	41.75	26.17	37.3	0.5580	98.7
Hadronization, multiplicity and fragmentation (IRC-unsafe)					
M	31.10	49.85	−60.3	1.005	96.8
n_{ch}	25.58	38.38	−50.1	0.7118	97.2
f_{ch}	4.095	3.766	8.0	0.4480	89.1
p_T^D	33.05	16.05	51.4	1.121	96.6
q	7.982	3.921	50.9	0.3856	95.2
[†] IBU's purity binning produced a single bin, so IBU failed to unfold.					

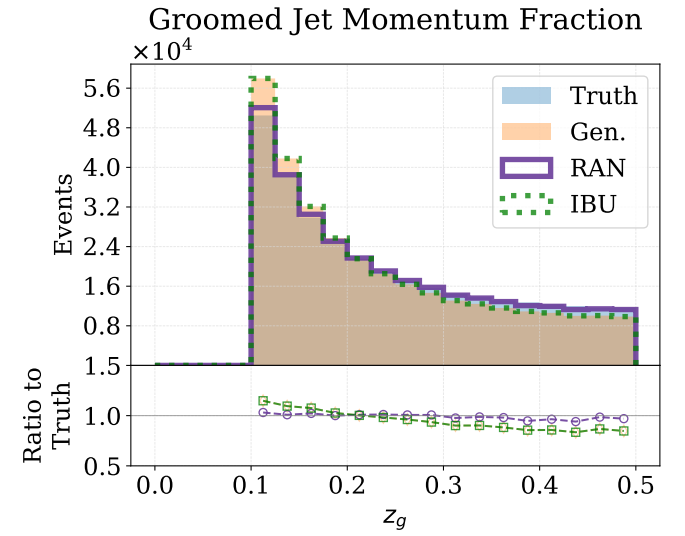
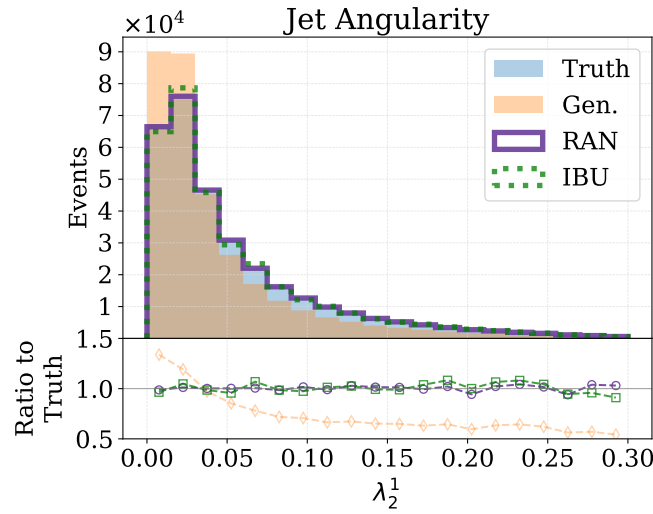
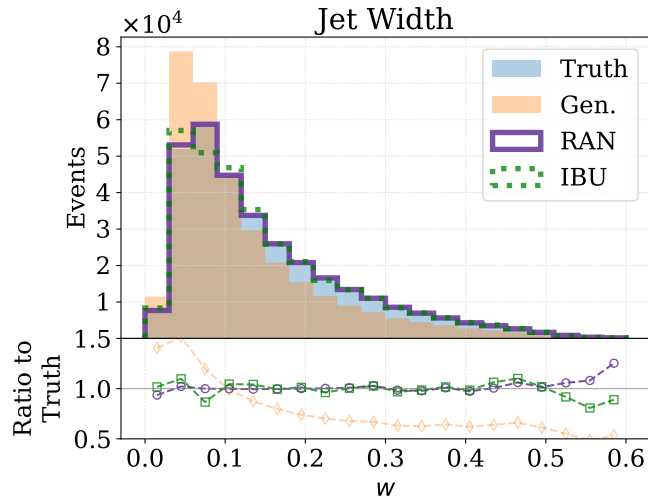
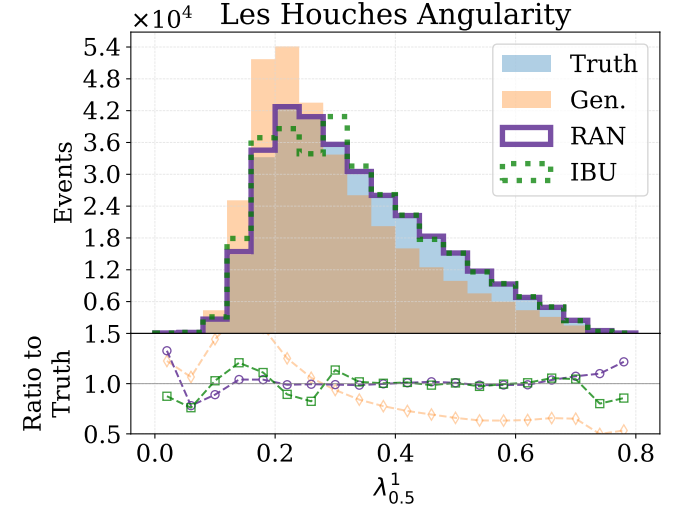
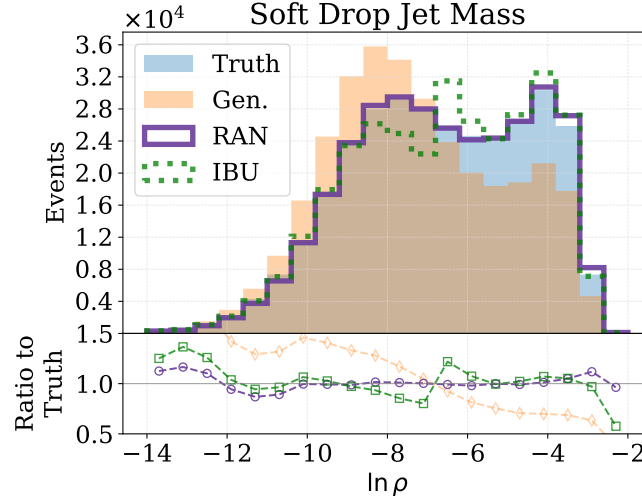
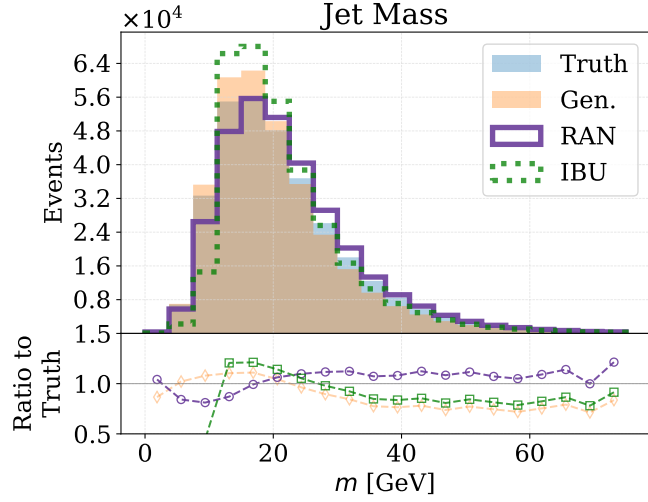
Detector Level (1 of 2)



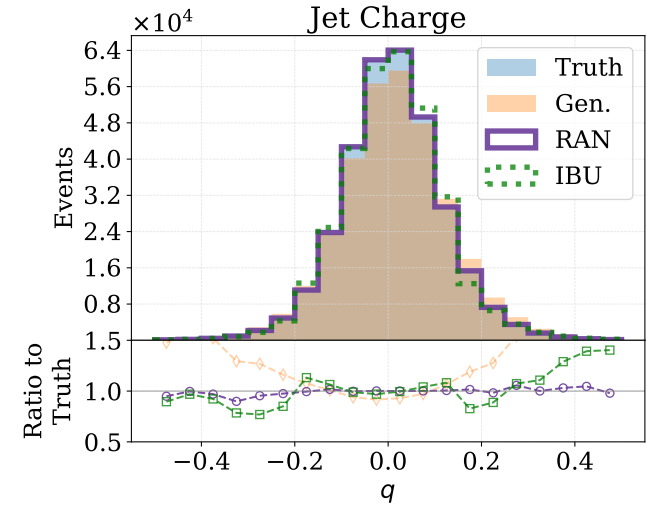
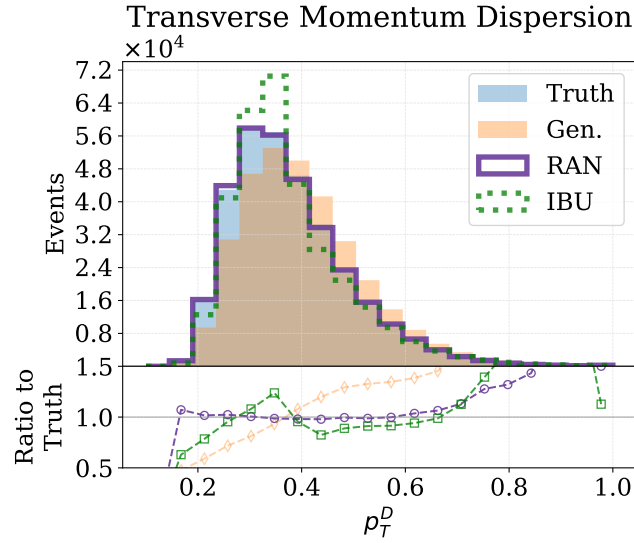
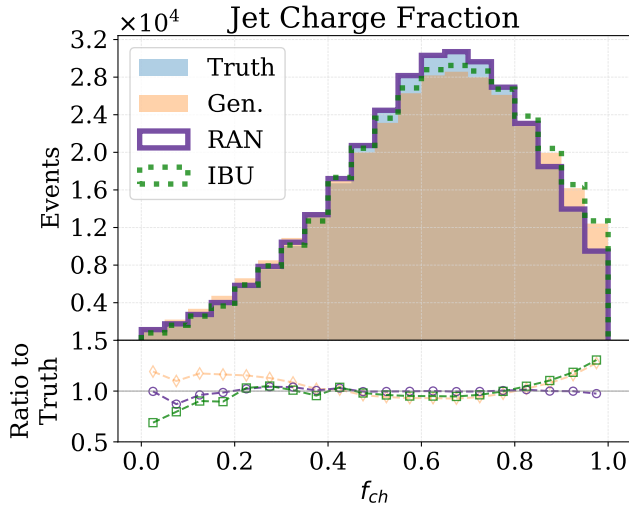
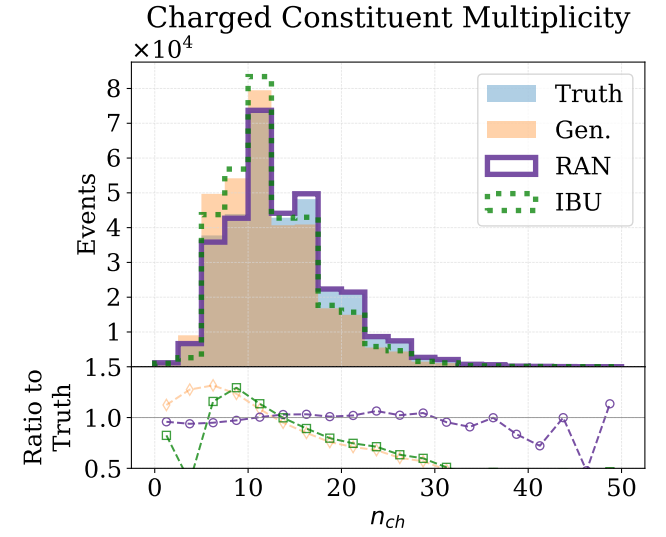
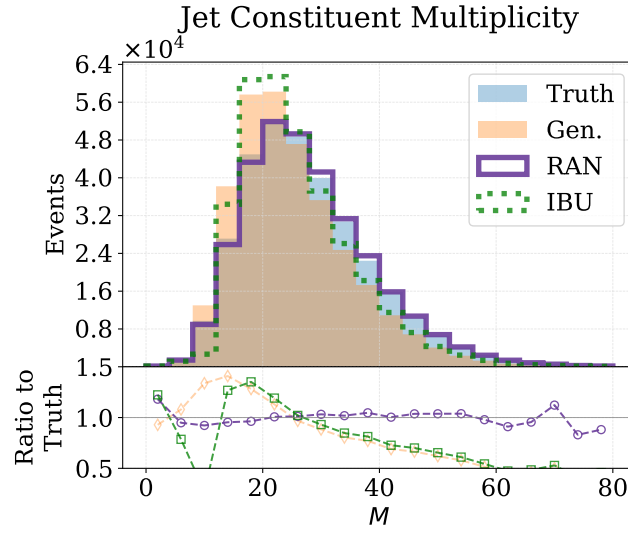
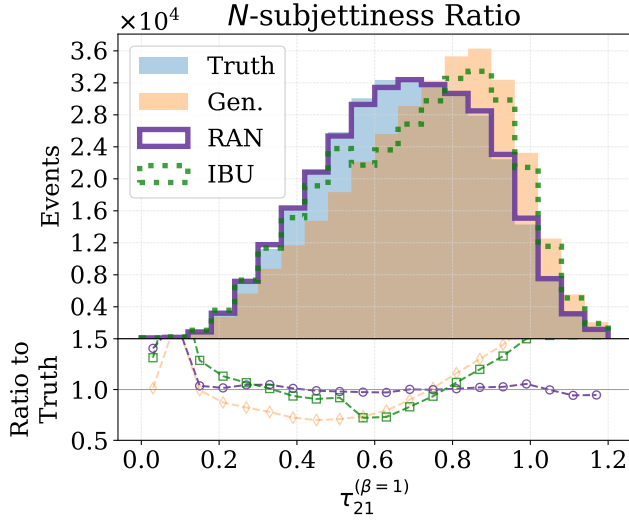
Detector Level (2 of 2)



Particle Level (1 of 2)

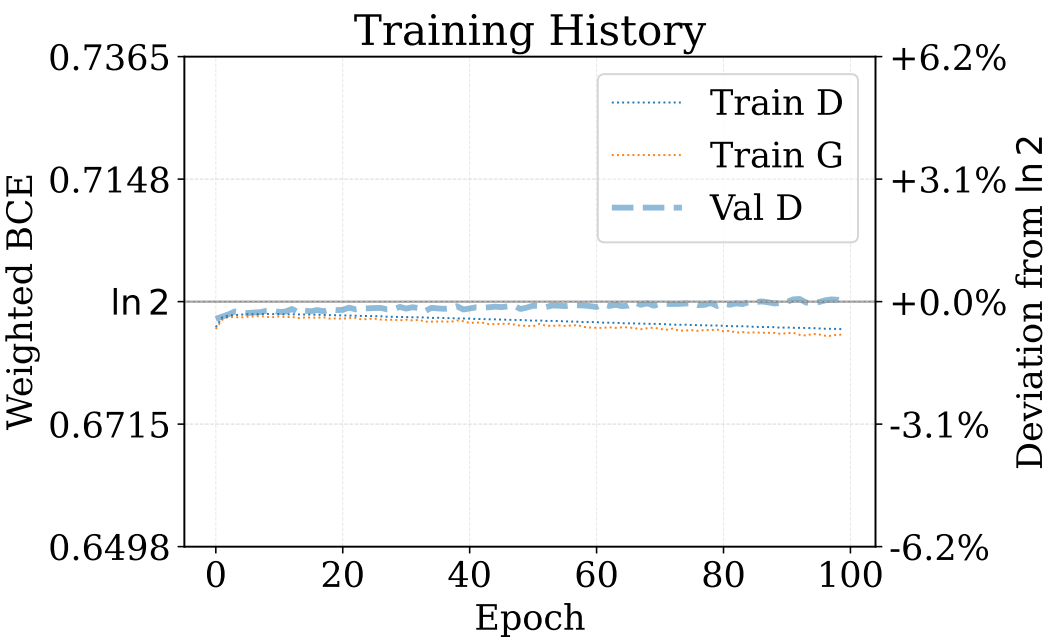


Particle Level (2 of 2)



Training diagnostics

Training losses



Selection criteria

